Sex and Love

Sexual reproduction is fundamental to passing on our genes. Multiple emotions evolved that accompany and facilitate this process, which together can create the feeling of love.

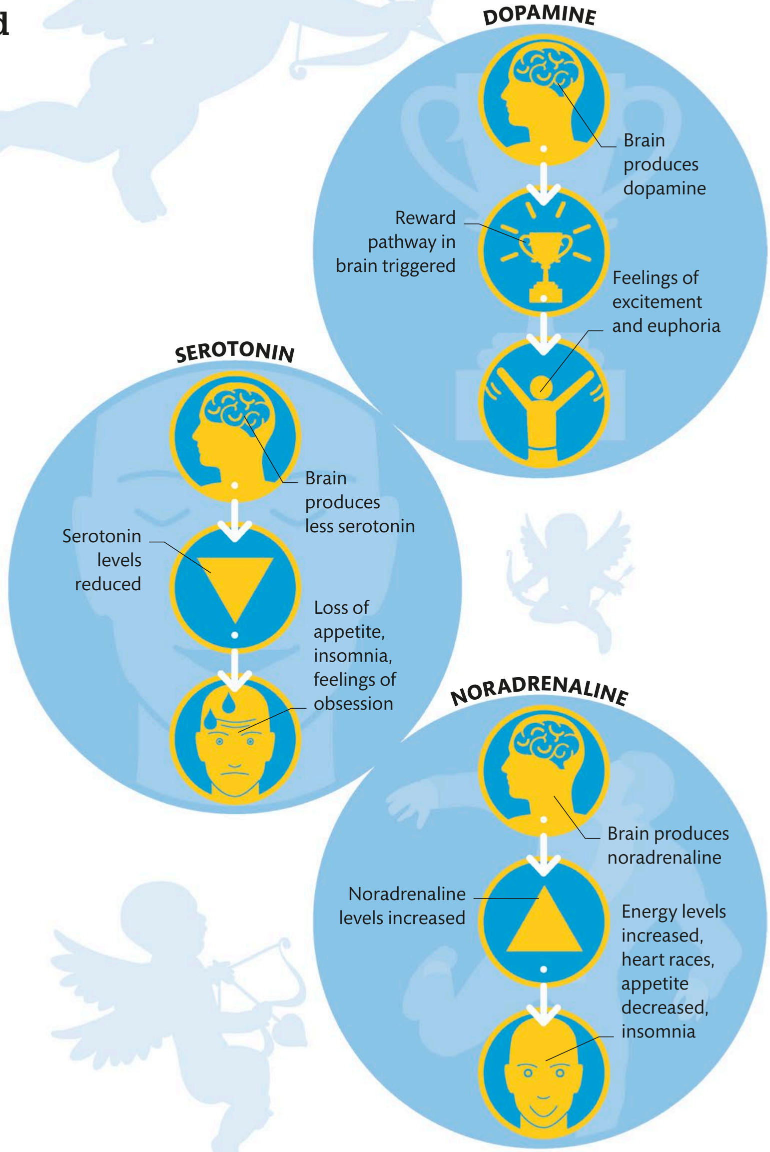
Love and attraction

The scientific study of love and sexual behavior has identified three primary components: attraction, attachment, and lust. These states all occur on different timescales and involve different regions of the brain producing an array of chemical messengers—neurotransmitters and hormones. Lust and attraction are closely interlinked, and both are transient, passing in a relatively short time. For relationships to last, these states must yield profound attachment, which involves long-term changes to the brain.

KEY

- Prefrontal cortex
- Hypothalamus
- Pituitary gland

Brain areas
The hypothalamus and pituitary gland control early hormone-led phases of bonding. The prefrontal cortex then mediates the emotional control involved in attachment.



Attraction
Surges of the chemical messengers dopamine and noradrenaline combine with reduced levels of serotonin to produce urgent feelings of attraction. In an energized state—with racing heart, sweaty palms, and little appetite—we think constantly about our lover, craving their company.

THE LOVE DRUG

Oxytocin, which is released by the hypothalamus, has long been known as the hormone that induces labor in mammals. It was then found to be crucial for mother-offspring bonding and later to be central to forming long-term attachments in sexual and social relationships.